Optimisation of desirable attributes of mycelium and their fruiting bodies through protoplast isolation, fusion, and regeneration

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ABSTRACT

This project is aimed at creating an improved hybrid species between mycelium belonging to two separate mushroom species, and the final desired hybrid should possess improved biological efficacy, heat tolerance, financial viability, and a reduction in pollutant concentration like heavy metals (copper and zinc ions), phosphate ions, crude oil, and plastics. The mushroom species that were mainly focused on in this project were *Volvariella Volvacea, Pleurotus Ostreatus,* and *Ganoderma lingzhi*. The hybridization of protoplast was mainly carried out on *Volvariella Volvacea* and *Pleurotus Ostreatus* because *Volvariella Volvacea* has increased heat tolerance while *Pleurotus Ostreatus* has shown to be effective in reducing heavy metal and phosphate concentrations. However, despite our valiant efforts, the protoplast hybridisation still proved to be unsuccessful despite a successful isolation of protoplast. PCR tests that were carried out to determine the species of the hybridised product were also inconclusive as it could not definitively confirm the successful hybridisation of both mushroom species. Many attempts at cultivating mushroom samples were unsuccessful too due to high rates of contamination.

INTRODUCTION

This research aims to create an improved somatic hybrid between dikaryotic mycelia as a means to increase biological efficacy, heat tolerance, financial viability as well as removal of pollutants (mycoremediation)[1] of mushroom species. Tropical regions comprising much of Southeast Asia, Africa and South America are home to some of the poorest nations on the planet[2]. Food is often lacking resulting in malnutrition and the development of Kwashiorkor disease in children[3]. Mushrooms have been identified as an excellent source of high quality protein that can be cultivated on agricultural waste such as sawdust, rice straw and sugarcane bagasse[4]. However, most mushroom species currently in commercial production are not suitable for cultivation in the tropics due to the excessively high ambient temperatures. Additionally, even though there are some mushroom species, such as the straw mushroom, that can be cultivated in the tropics, they have low yields and are difficult to market due to their extremely short shelf life. Therefore, through this research project, we aim to produce a somatic hybrid that both thrives in the tropical heat and is able to produce high yields. We hope that we will also be able to find a way to maximise mushroom production per unit area at minimum cost, providing an even cheaper source of food protein from agricultural and industrial organic wastes[5].

We plan to construct novel heat tolerant strains through Polyethylene glycol (PEG) induced protoplast fusion[6]. We are currently reviewing possible species to undergo protoplast fusion. Some possible species include Volvariella volvacea[7], Pleurotus eryngii[7], Pleurotus ostreatus[8],

Agaricus blazei[9], Agaricus bisporus[9], Hericium erinaceus[10], Auricularia auricula-judae, Lentinula edodes[11], Ganoderma lucidum[11], Lentinus sajor-caju[12], and Calocybe indica[13]. Heat tolerant hybrids will be isolated by exposing the hybrids to the lethal time of the parent strains at 40°C. The remaining surviving hybrids will therefore have a greater heat tolerance than their parent strains. Small scale trials will be undertaken to determine the biological efficacy which is calculated by the ratio of the mass of the mushrooms produced over the initial mass of the dry substrate.

MATERIALS AND METHODS

1) Subculture and Obtaining of Pure Strains of mycelium

Samples were procured from online sources. They were then cultivated in an agar medium at 28°C. The growth of new hyphae (a long, branching, filamentous structure of a fungus) was observed, and when it becomes evident growing out of the transferred mushroom pieces without any contamination, it is then sub-cultured onto new growth medium under aseptic conditions. Store the obtained mother cultures at 4°C (39°F) and sub-culture every 6–12 months.

2) Protoplast Isolation

A mass of mycelia was incubated in 30ml of Sabouraud broth for 72 hours. The mixture was subsequently centrifuged at 11000 rotations per minute (RPM) for 8 min to isolate the mycelia mass. The supernatant was removed and rinsed with different osmotic stabilisers, including 0.684M NaCl, 0.6M KCl, Sabouraud broth and phosphate buffered saline (PBS). 0.1M phosphate buffer was added to all osmotic stabilisers except PBS. The mixture was centrifuged and the pellet incubated with 1ml of digestion mixture at 32°C for 5 hours. Digestion mixture constituted 20mg/ml of lysozyme with 0.9% sucrose solution. The mixture containing protoplasts and cell wall debris was harvested after 5 hours and filtered through a cell strainer to remove undigested material. The filtrate was centrifuged at 13 300 RPM for 5 min and the supernatant was removed. The pellet was rinsed twice and resuspended with the respective osmotic stabilisers.

3) Protoplast viability

0.1ml of each solution of resuspended mushroom protoplasts were added to 0.1ml of sterile water and left to incubate at room temperature for 30 minutes. Each water-mushroom protoplasts mixture was then added to 3ml of Sabouraud. 0.1ml of each solution of resuspended mushroom protoplasts (with no water added) was also added to separate tubes of 3ml of Sabouraud. All tubes were left to incubate at room temperature for 72 hours.

4) Protoplast fusion

A solution of 40% w/v PEG of molecular weight 6000 and 0.04M $CaCl_2$ was prepared. The protoplast was added to 1ml of PEG solution and left in the orbital incubator at 30°C. After 30 minutes, the sample was centrifuged to remove the supernatant. The resulting mycelial mass was then transferred and grown in liquid sabouraud dextrose broth with KCl osmotic stabiliser.

5) Phosphate test through Malachite Green assay

Obtaining Standard Curve

1M (136.09mg/ml) of potassium dihydrogen phosphate, KH_2PO_4 solution (phosphate solution) was prepared by dissolving 54.43g of KH_2PO_4 powder to 400ml of deionised water. Using the phosphate solution, 6 solutions of 1ml of varying concentrations, 0.20µg/ml, 0.50µg/ml, 1.00µg/ml, 2.00µg/ml, 3.00µg/ml and 4.00µg/ml were prepared by diluting a small volume of the phosphate solution in deionised water. 1ml of deionised water was also prepared as a control. 30ml of 0.68M molybdate solution, 4.7ml of 18.2M concentrated sulfuric acid and 25ml of 2mM malachite green solution was mixed together to produce a working reagent. 20µl of working reagent was added to each of the solutions which were then incubated in the absence of light at room temperature for 30 minutes. The absorbance of all reaction mixtures were then read at 640 nm , with the 1ml of deionised water used as blank. A graph of absorbance of solution OD_{640}/nm against concentration of phosphate solution/µg ml^{-1} was plotted and a best fit line was drawn to obtain the standard curve.

Preparation of phosphate concentrations

Two concentrations of 400ppm (0.4mg/ml) and 1000ppm (1mg/ml) were chosen to test the efficacy of straw and oyster mushrooms in reducing soil phosphate levels (mycoremediation). These two concentrations were specifically chosen because the maximum phosphate concentration in soil for which healthy plant growth could be sustained is 330ppm, whereas soil that has a phosphate concentration above 1000ppm is unable to be treated or saved.

The two phosphate concentrations were prepared by diluting 1M of potassium dihydrogen phosphate (KH_2PO_4) with deionised water. After that, sabouraud broth was spiked to produce 400ppm and 1000ppm of phosphate solution respectively. Before adding the spiked sabouraud broth to the mushrooms, their initial concentration was first checked against the standard curve by adding 2.5µl of spiked sabouraud broth to 997.5µl of deionised water, a 400 times dilution. 20µl of working reagent was added to the resulting solution and incubated in the absence of light at room temperature for 30 minutes. Their absorbance was then read in the spectrophotometer at 640 nm, and checked against the standard curve obtained previously to determine the original concentration.

Test for mushroom efficacy in reducing phosphate levels

For each type of mushroom, a fixed mass of mycelium was added to 25ml of Sabouraud spiked with varying phosphate concentration, namely 0, 400, 1000 ppm phosphate concentration. The mycelium were then left to grow for 1 week.

6) <u>2,2-diphenyl-1-picrylhydrazyl (DPPH) Inhibition Assay</u>

0.1mM 2,2-diphenyl-1-picrylhydrazyl (DPPH) solution was prepared by dissolving 4mg of DPPH in 100ml of ethanol. Mushroom fruit was dried in the oven overnight. The dried mushroom was then grinded using a mortar and pestle, and the 0.5g of dried mushroom fruit powder was added to 10ml of ethanol. The mushroom and ethanol mixture was weighed, and then left in a boiling water bath for 10 minutes. Afterwards, ethanol was added to top up the mixture to its original mass. The mixture was thoroughly shaked, and 1ml of the solution was added to a microtube and centrifuged for 5 minutes at 13300 RPM. All the liquid was then extracted to form a mushroom-ethanol solution.

The following mixtures were then prepared:

- 3ml of ethanol (M1)
- 1.5ml of ethanol and 1.5ml of DPPH (M2)
- 1.5ml of mushroom-ethanol solution and 1.5ml of ethanol (M3)
- 1.5ml of mushroom-ethanol solution and 1.5ml of DPPH (M4)

All the reaction mixtures were then incubated in the absence of light at room temperature for 30 minutes. After 30 minutes, the absorbance of all reaction mixtures was read at 517 nm, with Mixture M2 taken as the control. The % radical scavenging activity (RSA) of the mushroom extracts was calculated using the following formula:

$$\frac{[M2\ absorbance\ OD_{517}\ with\ M1\ as\ blank] - [M4\ absorbance\ OD_{517}\ with\ M3\ as\ blank]}{[M2\ absorbance\ OD_{517}\ with\ M1\ as\ blank]} \times 100\%$$

7) <u>Polymerase Chain Reaction (PCR) Test</u>

DNA Extraction/Removing of PCR Inhibitor

To prepare the DNA sample for extraction, 3µl of proteinase-K was added to 500µl of Chelex (styrene-divinylbenzene co-polymer containing iminodiacetic acid groups). A sample of mycelial mass was extracted from mycelium plates and added to sterile water, where it was thoroughly mashed. 100µl of mycelium extract was added to the proteinase-K-Chelex mixture. After shaking the mixture thoroughly, the mixture was placed in a water bath of 55°C for 15 minutes and shaking the mixture every 5 minutes. The mixture was then removed from the water bath and placed into a boiling water bath for 10 minutes to denature proteinase-K in the mixture. The mixture was then centrifuged for 8 minutes at 13300 RPM to separate the Chelex beads and the beads were subsequently removed.

DNA Sample Concentration and Viability

The absorbance of the diluted DNA sample was read at 260 nm with deionised water used as a blank in a spectrophotometer to determine the concentration of undiluted DNA sample in µg/ml, given by the formula = [DNA sample absorbance OD_{260}] × dilution factor × 50µg/ml. The absorbance of the sample was then read at 280 nm again with deionised water used as a blank. The ratio of absorbance at 260 nm to absorbance at 280 nm, given by the formula = [DNA sample absorbance OD_{260}] ÷ [DNA sample absorbance OD_{280}], was to be maintained at a value between 1.7 to 2.0.

Primers used

Specific nucleotides of each type of mushroom were first selected from the National Library of Medicine, National Centre for Biotechnology Information[14]. Afterwards, nucleotides sequences of primers were designed to target a specific portion of the mushrooms' nucleotides. Potential primers were first shortlisted using a software that was created to check that primers have guanine (G) and cytosine (C) content within a range of 45% to 60%, avoid repeated nucleotides of length 4 or more, ensure only 1 G or C nucleotide within the last 5 nucleotides in the 3' end of the primer, ensuring presence of a GC clamp. (Appendix A). Primers were then selected based on its specificity, by comparing nucleotide sequences on the National Library of Medicine, National Centre for Biotechnology Information, Nucleotide BLAST[15]. For each type of mushroom, the gene and primer nucleotide sequences are listed in Appendix A.

Preparation of PCR testing sample

The primers were diluted down to a concentration of 10µM. Based on the concentration of DNA sample determined above, a suitable volume of DNA sample was extracted, ensuring the mass of DNA sample added is lower than 50ng. The DNA sample and 2.5µl of each diluted primer, forward and backward primer, was added into a PCR microtube. Afterwards, 0.5µl of Taq polymerase, 2µl of dNTP at 10mM and 10µl of PCR buffer were mixed together to form the master mix. 37.5µl of sterile water was added to get 50µl of master mix. 25µl of master mix was then added in each PCR microtube, and sterile water was used to top up the final volume to 50µl.

Thermal cycling

PCR microtubes are then added into a thermal cycler. Samples in PCR microtubes are left at 95.0°C for 5 minutes for the initial denaturation of the DNA template. Afterwards, a cycle begins where the PCR microtubes are left at 95.0°C for 1 minute to further denature the DNA template, then at 55.0°C for 15 seconds for annealing, and then at 72.0°C for lengthening of the strand. The cycle repeats itself 40 times in total. Afterwards, samples are left at 72.0°C for 10 minutes for the final lengthening. As this process runs for a long time, samples are held at 10.0°C by the thermal cycler indefinitely until they are retrieved, where they are stored in the freezer for preservation. (Appendix A)

Agarose Gel electrophoresis

An agarose gel of 1.5% agarose was heated. After it has melted, 3µl of gelred loading dye was mixed with 40ml of melted agarose gel. The mixture was then poured into a mould with a comb placed into the mould at one end. It was then allowed to cool to solidify into a matrix containing several wells, formed by the comb. After removing the comb, the gel matrix was completely submerged in 1X TAE buffer solution. 12.5µl of bromophenol blue (loading dye) was added into each PCR microtube. 10µl of DNA samples of oyster and straw mushroom with the loading dye was added to each of the wells. 10µl of a standard consisting of various DNA molecules of known sizes and loading dye was added to the leftmost well too. 130V of electricity was then run through the TAE buffer solution to allow the DNA samples to migrate along the gel. The electric current was turned off when the DNA samples migrated about ¾ across the gel.

To view and analyse the sample, the gel was removed and brought to a dark room and viewed under an ultraviolet (UV) light.

RESULTS

Table 1: Protoplast viability with different osmotic stabilisers

Mushroom type	Osmotic Stabiliser	Protoplast Viability
Oyster	NaCl	No
	KCl	Yes
	SB broth	No

	PBS	No
Straw	NaCl	No
	KCl	Yes
	SB broth	No
	PBS	No

Table 2: Hybrid viability with different osmotic stabilisers

Osmotic stabiliser	Hybrid viability
NaCl	No
KCl	No
SB broth	No
PBS	No

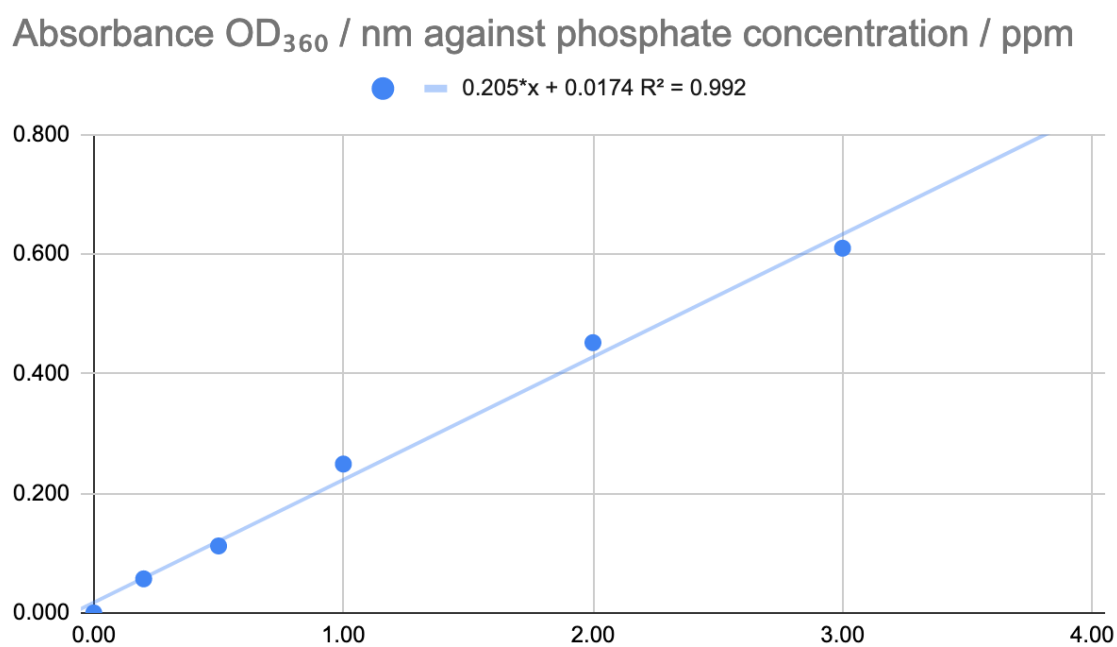

Figure 1: Phosphate concentration standard curve

The equation obtained was **y = 0.205x + 0.0174**, where x is the phosphate concentration / ppm and y is the absorbance OD_{360} / nm.

Table 3: Absorbance readings for 400ppm and 1000ppm phosphate concentration

Phosphate concentration	Absorbance OD_{360}	Estimated concentration
400ppm	0.018	1.17µg/ml (1.17ppm)
1000ppm	0.591	1119.2µg/ml (1119.2ppm)

Table 4: Absorbance readings for various mushroom species after phosphate test

Mushroom species	Absorbance OD_{360}	Estimated concentration
Straw	0.801	1528.8µg/ml (1528.8ppm)
Oyster	0.723	1376.8µg/ml (1376.8ppm)
Reishi	0.974	1866.4µg/ml (1866.4ppm)

Table 5: DPPH absorbance readings

Samples	Absorbance OD_{517} (Oyster)	Absorbance OD_{517} (Straw)
2ml ethanol (baseline)	-	-
1ml ethanol + 1ml DPPH	0.782	0.782
1ml oyster + 1ml ethanol (baseline)	-	-
1ml oyster + 1ml DPPH*	0.247	0.659
RSA	68.4%	15.7%

* rounded average absorbance taken from 3 samples

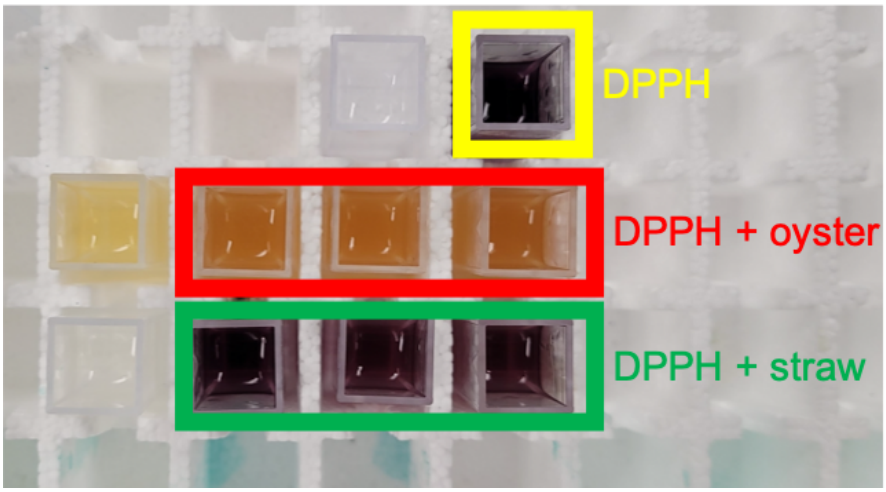

Figure 2: DPPH Assay Results

From the image, it can be seen that DPPH + oyster mushroom-ethanol solution led to a higher reduction in the purple colouration of DPPH, whereas in DPPH + straw mushroom-ethanol solution, the purple colouration mostly remained the same. This hence shows that the RSA of oyster mushroom is higher than that of straw mushroom.

Table 6: PCR Absorbance readings

	Straw	Oyster
Absorbance OD_{260} / nm	0.024	0.131
Absorbance OD_{280} / nm	0.004	0.159
Calculated concentration / µg ml^{-1}	60.0	327.5
Ratio OD_{260} / OD_{280}	6.0	0.82

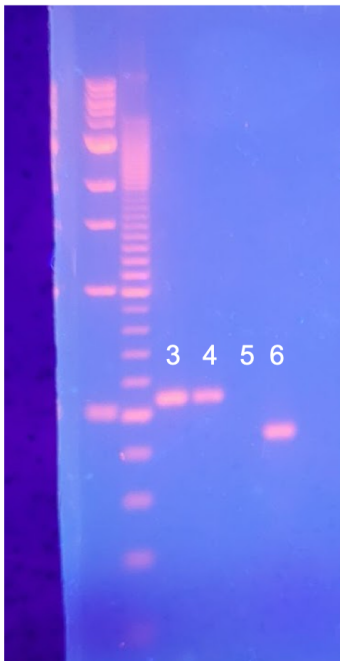

Figure 3: PCR gel electrophoresis results under UV

The 1st and 2nd lane from the left are the 1kb and 100bp ladder respectively while the 3rd and 4th lane consists of oyster mushroom mycelium and oyster mushroom fruit body respectively. As can be seen, the oyster gene identified was around 550 base pairs long, indicating a successful PCR test as the target gene was 508 base pairs long. (refer to Table 6 in Appendix A).
On the 6th lane (straw mushroom fruit body) from the left, the straw mushroom gene was also successfully identified, whose length was around 400 base pairs long, while the target gene was 373 base pairs long. However, the 5th lane (straw mushroom mycelium) had inconclusive results

due to the absence of DNA.

DISCUSSION

Our mushrooms experienced high rates of contamination with bacteria. This occurred regardless of their medium of cultivation (sabouraud broth, brown rice, sugarcane bagasse, or cocopeat), and regardless of their place of cultivation (cultivation room or microwave). After careful analysis, the possible reasons for contamination were faulty equipment like the flow hood cabinet, lack of sterile techniques, and sub-par laboratory skills. To address these issues, more research and background work could be done on proper sterile techniques and practice could even be carried out beforehand. Regarding our laboratory skills, the same logic applies as more practice could be carried prior to the experiment to hone our laboratory expertise. Last but not least, more funding could be pumped into the school to make investments in higher-quality equipment as existing equipment in the laboratory is due for an upgrade, like the spectrophotometer for example.

Cultures could be purchased from more reliable sources to ensure their viability. Additionally, the engagement of help from people equipped with greater expertise and skills could help to increase chances of our experiments turning out successful.

Isolation of the protoplast was successful only with KCl osmotic stabiliser and not with other osmotic stabilisers. Our hypothesis behind this trend is that the other osmotic stabilisers failed because of the lack of essential elements. However, protoplast fusion with KCl used as the osmotic stabiliser was unsuccessful, which is possibly due to the instability of the cell in the fusion mixture containing $CaCl_2$.

In our experiment, we attempted PCR tests of oyster and straw. However, it was only after multiple unsuccessful attempts that our PCR finally succeeded. Our previous PCR tests were unable to confirm the species of the mycelium as the genes amplified were only 50 to 100 base pairs long, showing that the primers did not bind to the DNA template accurately. This was due to a few possible reasons: incompatible primers, incorrect annealing temperatures, poor DNA samples, and inadequate micropipetting skills. Furthermore, the PCR tests carried out on straw mycelium were consistently unsuccessful because insufficient time was allotted to the DNA extraction process to break down the thicker-cell wall of the straw mycelium.

Oyster mushroom had a RSA of 68.4%. Straw instead, had a lower RSA than oyster at 15.7%. This shows that oyster mushrooms are better than straw at reducing free radical activity and thus reduce the risk of central nervous system diseases, cardiovascular diseases, and autoimmune and inflammatory disorders, greatly benefiting our health. This also proves that oyster mushrooms play a greater role than straw mushrooms in reducing the risk of contracting cancers too as free radicals contribute cell and DNA damage too.

CONCLUSIONS

In conclusion, the hybridisation of protoplast was inconsequential despite successfully isolating the desired protoplast. Although the PCR carried out on oyster mushrooms was successful, the

PCR carried out on the straw-oyster hybrid could not definitively confirm the identity of the possibly hybridised species.

Future work includes improving the procedure for protoplast fusion, confirming the identity of the mushroom, testing on a larger range of mushroom species and possibly trikaryotic protoplast fusion. A different method of fusing protoplast could be explored too, like using electricity to fuse cells (electrofusion).

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APPENDIX A

```
# parameters
mode = int(input("INPUT MODE [1: forward, 2: reverse]: "))
length = 20
gc_clamp = 1
max_repitition = 4
k = 5
acceptable_gc = [1]
print("Sequence: ")
sequence = ""
# input sequence
temp = 0
while (temp != "-1"): # terminates on -1
    temp = input()
    sequence += temp.strip()
nt = {'G': 'C', 'C': 'G', 'A': 'T', 'T': 'A'}
# prevents repetition of nucleotides
def repitition_func(seq, max_repitition):
    if max_repitition <= 0:
        return True
    else:
        if 'A'*max_repitition in seq or 'C'*max_repitition in seq or
'T'*max_repitition in seq or 'G'*max_repitition in seq:
            return False
        else:
            return True
# only preserve characters that are A, C, G or T
sequence = sequence.strip().upper()
sequence = ''.join([i for i in sequence if i in 'ACGT'])
final = ""
for i in range(length-1, len(sequence)):
    seq = sequence[i-length+1:i+1]
    # reverses sequence and inverts nucleotides for reverse primer mode
    if mode == 2:
        seq = ''.join([nt[i] for i in seq[::-1]])
    # ensure 1 G/C within last 5 nucleotide for 3' end
    ending_gc = seq[-k:].count('G') + seq[-k:].count('C')
    # calculate proportion of G/C in whole sequence
    whole_gc = (seq.count('G') + seq.count('C'))/length
    # prints acceptable sequences only
    conditions = ending_gc in acceptable_gc and 0.45<=whole_gc<=0.6 and ((not
gc_clamp) or (gc_clamp and (seq[-1] == 'G' or seq[-1] == 'C'))) and
repitition_func(seq, max_repitition)
    if conditions:
        final += ("{seq} {pos} {gc}\n".format(seq=seq, pos=str(i-length+2),
gc=whole_gc))
print(final)
```

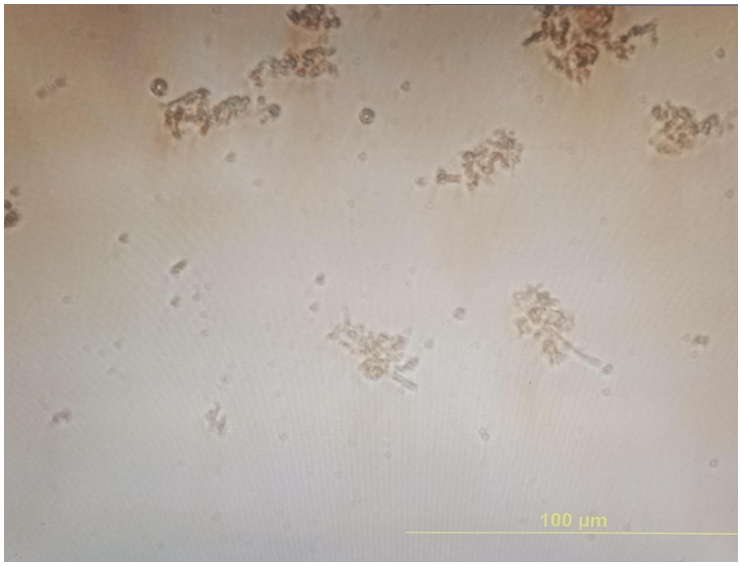

Figure 4: Isolated protoplast for volvariella volvacea

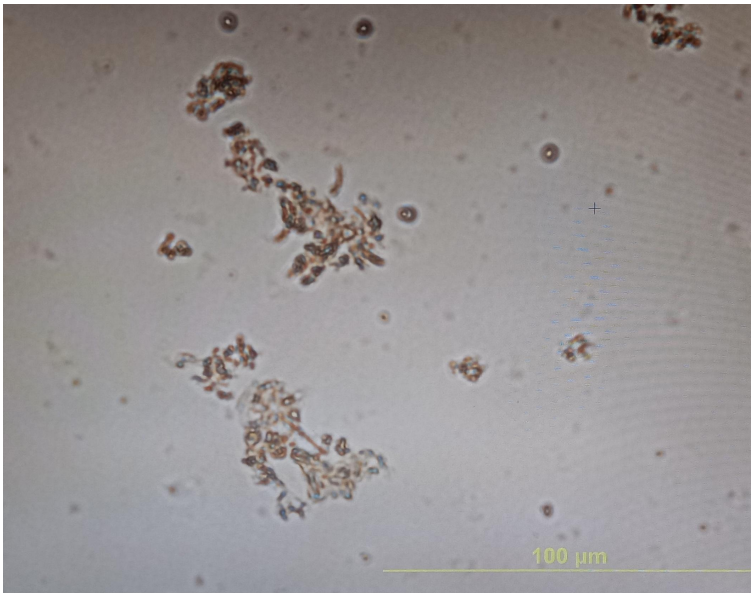

Figure 5: Isolated protoplast for pleurotus ostreatus

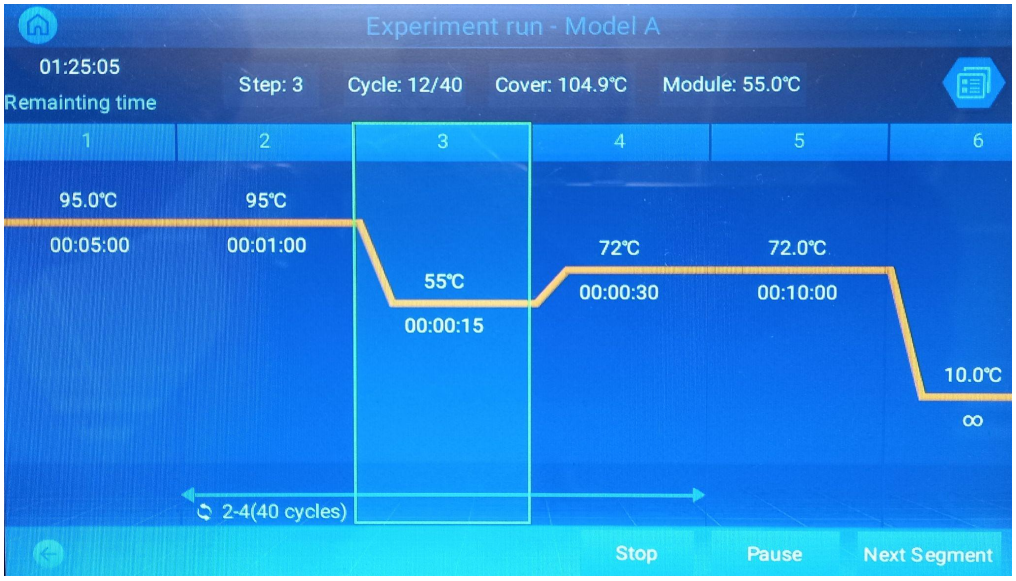

Figure 6: Thermocycler settings for PCR

Table 7: Gene sequences of primers and DNA strand length for both mushrooms

Nucleotide Sequence	Type	Melting Temperature / °C
Volvariella volvacea		
Volvariella volvacea 1-aminocyclopropane-1-carboxylate synthase 3 gene (MG582590.1)		
TCTGCAGGGA GTGAGTAATC	Forward	53.7
CTCGCTGGGA ATGGATAATC	Reverse	52.9
Target DNA strand length: 373		
Pleurotus ostreatus		
orotidine-5'-monophosphate decarboxylase (pyrG) gene (JF795324.1)		
AGCGTGGATG TGACGAAATC	Forward	55.4
TGCCCAATGA ACCCGATTAC	Reverse	55.4
Target DNA strand length: 508		